Chapter 6: Statistics - Question Bank Extraction

Question 1 (A): Multiple Choice Questions [1 Mark Each]

Q1.

Following table shows the percentage of donors as per the blood group, the central angle for the group 'O' is:

Blood group	O	A	B	AB
% Of persons	60	20	15	5

- (A) 72°
- (B) 108°
- (C) 216°
- (D) 54°

Q2.

If the following information regarding the budget of a family is shown as a pie graph, then the central angle for clothing is:

Head	Rent	Clothing	Education	Saving	Food	Miscellaneous
Expenditure	₹ 2400	₹ 1800	₹ 1200	₹ 1200	₹ 4800	₹ 600

- (A) 72°
- (B) 54°
- (C) 36°
- (D) 18°

Q3.

The class mark of the class $15 - 20$ is _____.

- (A) 17.5
- (B) 15.5
- (C) 17
- (D) 18.5

Q4.

If assumed mean $(A) = 57.5$, $\sum f_i d_i = -10$, and $\sum f_i = 80$, then $\text{Mean} =$ _____.

- (A) 57.625
- (B) 57.125
- (C) 57.375
- (D) 57.275

Q5.

Which of the following is not a measure of central tendency?

- (A) Median
- (B) Deviation
- (C) Mode
- (D) Mean

Q6.

If the mean of a frequency distribution is 34.9, $\sum f_i x_i = 1000 + a$, and $\sum f_i = 30$, then the value of $a =$ _____.

- (A) 47
- (B) 48
- (C) 42
- (D) 45

Q7.

A box contains nails of different lengths as shown below, the median class will be:

Length (in cm)	2.0–2.5	2.5–3.0	3.0–3.5	3.5–4.0	4.0–4.5
No. of nails	5	18	7	11	9

- (A) $2.5 - 3.0$
- (B) $3.0 - 3.5$
- (C) $3.5 - 4.0$
- (D) $4.0 - 4.5$

Q8.

Look at the pie diagram showing the number of skilled workers in different fields. The number of workers in production is ---------- if the total number of workers is $10,000$.

- (A) 2000
- (B) 1500
- (C) 2500
- (D) 3500

Q9.

Class marks in a grouped frequency table are useful to find:

- (A) Mean
- (B) Median
- (C) Mode
- (D) All of these

Q10.

The given data is to be shown by a frequency polygon. The coordinates of the point to show the number of students in the class $130 - 140$ are:

Heights (in cm)	100–110	110–120	120–130	130–140	140–150
No. of students	6	8	14	10	4

- (A) $(130, 10)$
- (B) $(135, 10)$
- (C) $(10, 135)$
- (D) $(10, 130)$

Question 1 (B): Short Answer Questions [1 Mark Each]

Q1.

Find the class mark of the class $55 - 60$.

Q2.

Write the Modal class for the given frequency distribution:

Class interval	10–15	15–20	20–25	25–30	30–35
Frequency	30	35	40	75	15

Q3.

Marks of 60 students in a class are tabulated below. For finding mode, state the Lower limit of the modal class $(L) =$ _____ .

Marks	10–19	20–29	30–39	40–49	50–59	60–69
Number of students	3	8	16	12	10	8

Q4.

Observe an adjacent pie diagram showing percentages of vehicles passing a signal in a town between 8 am and 10 am. Find the central angle for cars.

Q5.

The table given below gives the marks of 160 students in a school:

Marks	20–30	30–40	40–50	50–60	60–70	70–80	80–90	90–100
No. of students	8	10	15	25	30	38	24	10

How many students obtained marks above 80 ?

Q6.

If the mean of the frequency distribution is 8.1 and $\sum f_i x_i = 132 + 5k$, $\sum f_i = 20$, then find the value of k.

Question 2 (A): Activity Based Questions [2 Marks Each]

Q1.

Observe the frequency polygon data and complete the following table:

Class	18–19	19–20	20–21	21–22	22–23	23–24
Class mark	18.5	□	20.5	21.5	□	23.5
Frequency	4	13	15	19	17	6
Coordinates of points	(18.5, 4)	(19.5, 13)	□	(21.5, 19)	□	(23.5, 6)

Q2.

Complete the following activity to find the mean:

Weekly wages (Rupees)	Class mark (xi)	No. of workers (fi)	fixi
0 –	1000	15	15000
2000 –	□	35	105000
4000 –	5000	50	250000
6000 –	7000	20	140000
Total		$N =$	$\sum f_i x_i =$

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{\square}{120} = \square$$

Q3.

The following table shows causes of noise pollution. Complete the table to draw a pie diagram:

Causes	Percentage	Central angle
Construction	10%	□
Traffic	□	180°
Aircraft take off	9%	□
Industry	20%	□
Trains	11%	39.6°

Q4.

Complete the following activity to find the median:

Class (Student's marks)	No. of students (fi)	Cumulative frequency

		less than the upper limit (cf)
$0-$	4	4
$20-$	20	24
$40-$	30	54
$60-$	40	94
$80-$	6	100

Here, $L = \square$, $N = 100$, $\frac{N}{2} = 50$, $cf = 24$, $f = 30$, $h = 20$.

$$\text{Median} = \square \quad \text{...(Formula)}$$

$$\text{Median} = \square$$

$$\text{Median marks} = \square$$

Question

2 (B): Short Answer Problems [2 Marks Each]

Q1.

For the table given below, prepare a cumulative frequency table (less than type):

Daily No. of hours	8–10	10–12	12–14	14–16
Number of workers	150	500	300	50

Q2.

Find the mode for the following frequency distribution:

Class	1–10	11–20	21–30	31–40	41–50
Frequency	2	3	5	7	1

$$\text{Use formula: Mode} = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right] \times h$$

Q3.

Find the mean of the given distribution:

Difference in ages (in years)	No. of couples (fi)
$0-$	1
$2-$	2

$4-$	8
$6-$	5
$8-$	3
$10-$	1

Question 3 (A): Activity Based Questions [3 Marks Each]

Q1.

The following table shows the funds collected by 50 students for flood affected people. Find the mean of the funds.

Fund (Rupees)	0-1000	1000-1500	1500-2000	2000-3000
No. of students	6	24	18	2

Activity Solution: Let $A = 1250$, examining all d_i , class size ratio factor $g = 250$.

Class Fund (₹)	Class mark xi	di=xi-A=xi -1250	ui=gdi	Frequency fi	fiui
$0-$	500	$\square$	-3	6	-18
$1000-$	$1250 \rightarrow$	0	0	24	0
$1500-$	1750	500	2	18	$\square$
$2000-$	2500	1250	$\square$	2	10
Total				$\sum f_i =$	$\sum f_i u_i =$

$$\bar{u} = \frac{\sum f_i u_i}{\sum f_i} = \frac{28}{50} = \square$$

$$\bar{u} \cdot g = \square \times 250 = \square$$

$$\bar{x} = A + g \cdot \bar{u} = 1250 + 140 = 1390$$

$\therefore$ The average of the funds collected is ₹ 1390 .

Q2.

Complete the activity to find the mean by the direct method:

Weekly wages (Rupees)	Class mark (xi)	No. of workers (fi)	xifi

1000 –	1500	25	37500
2000 –	□	45	112500
3000 –	3500	50	□
4000 –	4500	30	□
Total		$N =$	$\sum f_i x_i =$

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{\square}{150} = \square$$

Q3.

Complete the following activity to find the median marks of the students:

Class (Student's marks)	No. of students (fi)	Cumulative frequency less than upper limit (cf)
0 –	4	4
20 –	20	24
40 –	30	54
60 –	40	94
80 –	6	100

Here, $L = \square$, $N = 100$, $\frac{N}{2} = 50$, $cf = \square$, $f = 30$, $h = 20$.

$$\text{Median} = L + \left[\frac{\frac{N}{2} - cf}{f}\right] \times h \quad \text{...(Formula)}$$

$$\text{Median} = \square \quad \text{...(Substitute values)}$$

$$\text{Median} = \square + \frac{\square}{30} \times 20$$

$$\text{Median marks} = \square$$

Q4.

The table below gives the ages of 120 drivers of cars involved in accidents during a certain year. Complete the table and draw a pie diagram:

Age of driver (in years)	Number of drivers	Central angle
Under 20	25	75°
20 –	50	□
40 –	35	105°

Over 60	10	30°

Question 3 (B): Long Answer Questions [3 Marks Each]

Q1.

Draw a histogram for the following data:

Wages (in ₹)	40–50	50–60	60–70	70–80	80–90
No. of workers	3	8	12	6	4

Q2.

In a school, the weekly pocket money of 50 students is given below. Find the mode of the pocket money:

Pocket money (in ₹)	40–50	50–60	60–70	70–80	80–90	90–100
No. of students	2	8	12	14	8	6

Q3.

Draw a frequency polygon for the following distribution:

Weight of packet	0–5	5–10	10–15	15–20	20–25
Number of packets	3	5	8	5	4

Question 4: Advanced Application Problems [4 Marks Each]

Q1.

Find the mean by step-deviation method:

Class	10–15	15–20	20–25	25–30	30–35	35–40
Frequency	5	6	8	12	6	3

Q2.

If the mean weight of a packet is $12.9\ \text{gm}$, find the missing value of b:

Weight of packet	0–5	5–10	10–15	15–20	20–25
Number of packets	3	b	8	5	4

Q3.

Find the mode for the following distribution:

Ages (in years)	1–10	11–20	21–30	31–40	41–50
No. of children	2	3	5	7	1

Q4.

Calculate the missing frequency from the following distribution, given that the median of the distribution is 24:

Age in years	0–10	10–20	20–30	30–40	40–50
No. of persons	5	25	?	18	7

Q5.

Draw both a histogram and a frequency polygon on the same graph for the following data:

Max. temp (◦C)	24–28	28–32	32–36	36–40	40–44
No. of towns	4	5	7	8	6

Question 5: Creative & Higher Order Questions [3 Marks Each]

Q1.

The angles of a triangle $\triangle ABC$ are in the ratio $1:2:3$. Represent this information in a pie diagram.

Q2.

120 students of Standard 10 were surveyed about their favorite sports and a pie chart was prepared. Observe the pie diagram and answer the following questions:

1. How many students like to play Tennis?
2. What will be the central angle for Kho-Kho?
3. How many students like Kabaddi?

Q3.

Observe the given frequency polygon. Prepare the corresponding grouped frequency table and find the mean of the Net Asset Value (NAV).